

Karim Mansour

Machine Learning Engineer

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SUMMARY

Six years building NLP systems that run in production rather than in notebooks. My work is usually the unglamorous half: the evaluation harness, the chunking strategy, the reason p99 is 400ms instead of four seconds.

EXPERIENCE

Machine Learning Engineer, Meridian Document AI - Remote

04/2022 - Present

- Own the retrieval layer for a contract analysis product indexing 42 million document chunks in pgvector, using HNSW indexes with $m=16$ and $ef_construction=64$.
- Built the offline evaluation set of 3,100 labelled query-document pairs and the harness that reports $precision@5$, $recall@20$ and $nDCG@10$ on every model change. Before it existed, retrieval changes were argued from anecdotes.
- Fine-tuned a bge-base embedding model on in-domain contract language with a contrastive objective in PyTorch, improving $nDCG@10$ from 0.61 to 0.74 against that set.
- Replaced fixed-size chunking with a layout-aware splitter that respects clause boundaries; $recall@20$ rose 9 points with no change to the model.
- Cut serving cost 62 percent by quantising the reranker to int8 with ONNX Runtime and batching requests, holding p99 latency under 380ms.
- Built the RAG pipeline that grounds every generated answer in a retrieved span, including the span-verification step that rejects an answer whose quote is not present in the source.

Data Scientist, Nour Analytics - Beirut

08/2019 - 03/2022

- Trained text classification models for Arabic and English support tickets, reaching macro F1 of 0.88 across 14 intent classes.
- Moved the team off ad hoc scripts onto MLflow for experiment tracking and model registry.
- Served models behind a FastAPI service on Kubernetes with canary rollout by traffic percentage.

EDUCATION

MSc Data Science, Ecole Polytechnique, 2019

BSc Computer Science, Lebanese American University, 2017

SKILLS

Python, PyTorch, Hugging Face Transformers, sentence-transformers, ONNX Runtime, FAISS, pgvector, MLflow, FastAPI, Docker, Kubernetes, SQL, NumPy, pandas.

OPEN SOURCE

Maintainer of a small library for evaluating retrieval pipelines; roughly 700 stars.